

Jinil Kim

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RESEARCH INTEREST

- Identifying computational signatures in human brain and behavior
- Translating computational signatures into personalized interventions and better AI learning systems

EDUCATION

Seoul National University

B.A. Linguistics · B.A. Psychology · B.S. Computer Science

GPA: 4.18 / 4.3 (3.96 / 4.0) | Major GPA (4.0): Ling./Psych.: 4.00, CS: 3.82

* Mandatory military service, Jul 2023 – Jan 2025

Seoul, South Korea

Mar 2021 – Exp. Feb 2027

The University of Hong Kong

Summer Research Program, Faculty of Psychology

Hong Kong SAR

Jun 2025 – Aug 2025

RESEARCH EXPERIENCE

Mental health, Internet, Neuroscience & Decision-making Lab

Visiting Researcher (Advisor: Prof. Yuan-Wei Yao)

University of Hong Kong

Jun 2025 – Aug 2025

- Developed a time-varying Drift Diffusion Model (tDDM) to investigate the hierarchical influence of emotion and reward prediction errors in social decision-making

Computational Clinical Science Laboratory

Undergraduate RA (Advisor: Prof. Woo-Young Ahn)

Seoul National University

Apr – Jun 2025; Sep 2025 –

- Developed an Adversarial Inverse Reinforcement Learning (AIRL) model with interpretable parameters to capture individual differences in decision policies.
- Designed a hierarchical multimodal model integrating EEG, behavioral, and app-based digital phenotype data to predict alcohol craving.

ONGOING PROJECTS

Development of a Personalized Color Vision Correction Display Filter Using fMRI-Based Neural Responses and Deep Learning

Project Lead (Advisor: Prof. Jiook Cha)

Apr 2025 – Present

- Designed the full research pipeline: experiment design, data preprocessing, neural modeling, and optimization-based filter generation.
- Analyzed inter-individual coherence and disparity of individuals with color vision deficiencies in neural color geometry using Procrustes alignment.
- Funded by SNU Undergraduate Research Grant (\$3,500).

Modeling Naturalistic Human Decision-Making with Bounded-Foresight Inverse Reinforcement Learning

Research Analyst (Advisor: Prof. Woo-Young Ahn)

Sep 2025 – Present

- Developing a planning-aware inverse reinforcement learning framework to infer latent planning depth from human sequential decision-making.

A Time-Varying Drift Diffusion Model of Emotion and Reward Prediction Errors in Depression

Project Lead (Advisor: Prof. Yuan-Wei Yao)

Jun 2025 – Present

- Developed a time-varying Drift Diffusion Model (tDDM) to investigate the hierarchical influence of emotion and reward prediction errors in social decision-making

CONFERENCES

- Kim, Jinil, Cho, A. M., Seo, J., Cha, J. (Jul 2026). *Inferring Individualized Color-Vision Distortions from fMRI Hue-Representation Geometry*. Accepted for presentation at the ICML 2026 Workshop on Structured Data for Health.
- Kim, Jinil, Jeong, J., Yun, H., Oh, M., Ahn, W.-Y. (Dec 2026). *Modeling Naturalistic Human Decision-Making with Bounded-Foresight Inverse Reinforcement Learning*. Under revision for NeurIPS 2026.
- Kim, Jinil, Cho, M., Seo, J., Cha, J. (Jun 2026). *fMRI Decoding Reveals Intact Neural Color Representations in Color Vision Deficiency*. Poster accepted for presentation at the 2026 Organization for Human Brain Mapping (OHBM) Annual Meeting, Bordeaux, France.

GRANTS

KOSAF Humanity 100 Years Undergraduate Scholarship Mar 2025 – Jul 2026
Amount: \$12,500 (KRW 17,000,000)

The University of Hong Kong Foundation for Educational Development and Research Scholarship Jun 2025 – Aug 2025
Amount: \$1,300 (HKD 10,000)

Superior Academic Performance scholarship Mar 2022, 2023; Sep 2022

HONORS & AWARDS

Best Poster Award in SNU Undergrad Research Program (4th Place) Jan 2026
SNU Liberal Arts Competition (1st Place) Jul 2023

TEACHING

Seoul National University Seoul, South Korea
Teaching Assistant, Basic Computing (English) Mar 2026
Teaching Assistant, Basic Computing (Korean) Sep 2025

SKILLS

Computational Modeling: Bayesian hierarchical models, Reinforcement Learning
Neuroimaging: fMRI preprocessing, Neural decoding
Programming: Python, R, Stan, C++, Java
ML & Data: Deep learning (MLP, CNN), high-dimensional time-series

LANGUAGES

Korean (Native), English (Full Professional Proficiency), French (Beginner)